UNIVERSITY OF BAYREUTH

Causal Inference Spring 26

Take-Home Exam

Due on 10 August at Noon

Word limit: 7000 words

Exercise 1: The Cause of Cholera Epidemics

In this exercise, you will carry out the diff-in-diff analysis implied by Table XII of John Snow's *On the Mode of Communication of Cholera* (1855), which is reproduced below.[1] Snow tried to establish that cholera was spreading through contaminated water. In other parts of the study, he presents evidence that nearby houses connected to different water supplies had different cholera risks, and that cholera outbreaks were concentrated around certain pumps providing public water. In this part he carries out an informal diff-in-diff, using the fact that the water source for some London sub-districts changed in 1852 when the Lambeth Company moved its intake from Hungerford Bridge (near Westminster) to Thames Ditton, 22 miles upstream. Snow argues that the cleaner water obtained in sub-districts served by the Lambeth Company after the change reduced the risk of death from cholera in these sub-districts.

1. Create a panel dataset based on the table. Each row should be a subdistrict-year.
2. Define the treatment and control groups, justifying your choice. Report the mean number of deaths in each group in the pre and post periods; use these means to calculate the effect of the Lambeth Company's move assuming a valid diff-in-diff design.
3. State verbally the *parallel trends assumption* that must be employed here. Is it likely to hold? Under what circumstances might it be problematically violated? What data would be useful to assess this assumption?

[1] The book is available on Google Books.

4. Estimate the diff-in-diff (point estimate and standard error) using the **group diff-in-diff formulation, i.e. interaction formulation**.

5. What may be wrong with the standard errors in the above analysis?

6. Estimate the diff-in-diff (point estimate and measures of uncertainty) using the **fixed effects formulation**.

7. How does your definition of treatment and control groups affect your results?

TABLE XII.

Sub-Districts.	Deaths from Cholera in 1849.	Deaths from Cholera in 1854.	Water Supply.
St. Saviour, Southwark	283	371	Southwark & Vauxhall Company only.
St. Olave	157	161	
St. John, Horsleydown	192	148	
St. James, Bermondsey	249	362	
St. Mary Magdalen	259	244	
Leather Market	226	237	
Rotherhithe*	352	282	
Wandsworth	97	59	
Battersea	111	171	
Putney	8	9	
Camberwell	235	240	
Peckham	92	174	
Christchurch, Southwark	256	113	Lambeth Company, and Southwark and Vauxhall Compy.
Kent Road	267	174	
Borough Road	312	270	
London Road	257	93	
Trinity, Newington	318	210	
St. Peter, Walworth	446	388	
St. Mary, Newington	143	92	
Waterloo Road (1st)	193	58	
Waterloo Road (2nd)	243	117	
Lambeth Church (1st)	215	49	
Lambeth Church (2nd)	544	193	
Kennington (1st)	187	303	
Kennington (2nd)	153	142	
Brixton	81	48	
Clapham	114	165	
St. George, Camberwell	176	132	
Norwood	2	10	Lambeth Company only.
Streatham	154	15	
Dulwich	1	—	
Sydenham	5	12	
First 12 sub-districts	2261	2458	Southwk. & Vauxhall.
Next 16 sub-districts	3905	2547	Both Companies.
Last 4 sub-districts	162	37	Lambeth Company.

Exercise 2: Assumptions & Research Design

In a recent but quite influential study, Ferwerda and Miller examine the effect of the Nazi administration rule on resistance. They use a Regression Discontinuity Design, based on the Vichy line: the line that was used to separate land occupied by Nazi Germany with the Vichy-governed French zone. Their findings led them to conclude that devolving governing authority significantly lowered levels of resistance: there are fewer resistance incidents in the Vichy area than in the Nazi occupied area, to the north of the line.

This finding has recently been questioned by Kocher and Monteiro, who criticize the Ferwerda and Miller study on various methodological grounds. Ferwerda and Miller subsequently responded to these criticisms.

Carefully read the debate around these studies and answer the following questions:

1. What is the main criticism by Kocher and Monteiro with regard to the utility of the Vichy line as an identification tool in this context? Use potential outcomes terminology and refer to the RD assumption in your answer.
2. What tests do Ferwerda and Miller implement to address this criticism?
3. Having read both sides, do you accept or do you reject the evidence provided by Ferwerda and Miller? Elaborate on your assessment.
4. Briefly elaborate on the general utility of the RD in this context and on the potential sources of caution that one might need to have in mind when implementing this method for geographic/political boundaries.

Exercise 3: Insurgents and (Civil) Wars

A key problem incumbents encounter in civil wars is lack of information to combat insurgency. Given that insurgents exploit information asymmetries at the local level, they can easily hide and become a difficult target for incumbents. In the absence of such information, incumbents often resort to indiscriminate violence, via large-scale reprisals against entire villages suspected to host insurgents. One such example of indiscriminate violence is aerial bombardment. Due to the nature of insurgency, bombing frequently occurs in and around settled areas, and consequently it tends to generate many civilian casualties. Using data from the Vietnam War, Kocher, Pepinsky and Kalyvas examine the effect of such bombings on Viet Cong support. In particular, they look at the impact of September 1969 bombings on hamlet control in December 1969.

The data comes from various sources. The United States compiled a gazetteer of South Vietnamese hamlets, identified their geographic coordinates, and conducted a census. District Senior Advisors (DSAs), Army officers ranking major or above, were assigned to complete detailed questionnaires, some on a monthly basis, others quarterly, for every village and hamlet in their zones of operation. DSAs, together with small American staffs, were detached from U.S. units to live and work in the districts they rated. The Republic of Viet Nam had 261 districts with a median area of 377 kilometres squared, or about one-fourth the size of the median U.S. county. There was a median of 36 hamlets per district in 1969. One might analogize the problem a DSA faced to that of the sheriff of a small U.S. county trying to identify dangerous towns or neighborhoods in his or her jurisdiction. Linking bombings to hamlets, the authors construct a dataset that allows them, within some margin of error, to identify the number of bombings per hamlet in September 1969 and to examine their impact on insurgency control. The variables of interest are as follows:

- `mod2a_1adec`: the "Enemy Military Model (2A)" (Hamlet Control) in December 1969, which rates the presence and activity of Viet Cong military units in the vicinity of each hamlet on a 5-point scale: "fully government controlled" (1), "moderately government controlled" (2), "contested" (3), "moderately insurgent controlled" (4), and "fully insurgent controlled" (5). While an ordinal variable, consider this variable to be continuous for the purposes of this exam.
- `bombed_969`: Number of bombings per hamlet in September 1969.
- `std`: Rough terrain
- `lnhpop`: log of hamlet population
- `ln_dist`: log distance from closest international border
- `score`: Development index score
- `mod2a_1ajul`: Enemy Military Model (2A) in July 1969
- `mod2a_1admn`: District average control before September 1969

Your task is as follows:

1. Discuss the suitability of matching and multivariate regression in the context of this research. What are their common assumptions? How do they differ?

2. Use a matching estimator to derive the effect of bombing in September 1969 on insurgency control in December 1969. To conduct the analysis, operationalise the treatment as a binary variable. Conduct the following steps:

    - Choose a matching estimator other than Caliper matching. Describe how you have chosen the estimator and upon which assumptions it rests.
    - Assess balance in pre-treatment covariates between treated and control units, before and after matching. Discuss which covariates you use in the matching process and why.
    - Estimate and interpret the effect.

3. Use Capiler matching using a capiler of 0.25 and estimate the treatment effect. Assess balance between the control group and the treated group before and after Caliper matching. Is Caliper matching doing a better job than the algorithm you chose in the previous question? Which matching procedure do you prefer and why?

4. Estimate the effect of the number of bombings in September 1969 on hamlet control in December 1969 using a multivariate regression by conditioning on covariates. Discuss which covariates from the dataset you have included and why. Does the point estimate and standard error you estimate differ from the ones you have found before? Discuss which result you find more convincing and why.

Analyses that rely on random number generators should include the following seed: set.seed(020260026).

Exercise 4: Institutions and Economic Development

This task is based on the famous Acemoglu, Johnson & Robinson (AJR) 2001 study on the importance of inclusive institutions on economic development.[2] AJR argue that institutions leave a long imprint on countries' economic activity. They distinguish between inclusive and extractive institutions. The

[2] Acemoglu, Johnson, & Robinson. 2001. "The Colonial Origins of Comparative Development: An Empirical Investigation." *American Economic Review*, 91(5): 1369-1401, available here.

former serve to diffuse economic returns along different strata, whereas the latter serve to appropriate wealth among few.

To find evidence for the importance of the distinction between good and bad institutions, they turn to the colonial structures of the 19th and early 20th century. Their identification strategy comes from the quasi-random variation in geography which determined whether colonizers would establish inclusive or extractive institutions. The logic is as follows: In those areas where settlers encountered high mortality rates, they built extractive institutions, without any long-term planning. In areas with low mortality rates, they built inclusive institutions. To the extent that the geographic determinants of mortality rates are as good as randomly assigned, they provide a very important instrument in order to test the economic returns of institutions.

The data can be found on the website as `AJR.dta`. The key variables are the following:

- `logpgp95`: the logged GDP per capita measured in 1995. (Logging GDP is a very typical way to bring its distribution closer to the normal, while changing the interpretation from levels to percentage). This is Y, the dependent variable of interest.
- `avexpr`: Average protection against expropriation risk (1985—1995). This is an index of how extractive a given set of institutions is. This is our D, which, of course, is not randomly assigned.
- `logem4`: Logged settler mortality rates. This is Z, our instrument.

You can also use other variables from the dataset if you think this helps in any part of your analysis. Your task is to answer the following questions:

- Is there a link between institutions and economic development? This is not a causal question, I'm just asking if there is any association between the two. Provide a scatterplot to show this is the case.
- Is this relationship causal? How do mortality rates help in answering this question?
- Provide a provide that reflects the reduced form equation. Estimate the ITT.
- Is the first-stage assumption valid in this case?
- Estimate the LATE, using both an - emulated - Wald estimator and a 2SLS estimator. Does conditioning on observables change the estimates (2SLS only)?

- After presenting their main results, the authors try to address criticisms about the validity of the instrument (practically about exclusion; read that section of the paper here) and briefly describe the authors' arguments. Now evaluate the exclusion restriction: Do you conclude the authors' IV design is valid? Why or why not?

Exercise 5: The Number of Candidates and Political Representation

In the aftermath of the 2016 US Presidential election, Jill Stein, the defeated candidate of the Green party, said in an interview that turnout would have been lower if she had not competed. When asked about the possibility that her candidacy contributed to the defeat of Hilary Clinton, Stein claimed that her supporters would have abstained if they could not have voted for her. This general intuition is shared by many: in countries in which political competition revolves around a few candidates like in the United States, the United Kingdom, or Canada, people whose preferences do not resonate with any of the candidates are more likely to abstain because they feel *alienated* by the system (for a review of the literature, see Blais, 2006). This is a strong case in favor of elections with more than two candidates, and flexible state regulation facilitating candidacy, as a way to increase turnout and reduce inequalities in political representation (Gallego, 2014). Several comparative and cross-sectional studies have investigated this question, but with mixed results: some find that the number of candidates decreases turnout (e.g. Jackman, 1987), some that it increases it (e.g. Taagepera et al, 2013), and yet others that it has no effect (e.g., Fornos et al, 2004).

Question a) Discuss why cross-sectional studies across countries, or across regions within a given country, could lead to mixed results. What type of selection bias could be present in the country cross-sectional studies and what type of selection bias could be present in within country regional cross-sectional studies? Write out a model specification you imagine they estimate and discuss sources of omitted variables bias using formal notation.

Question b) You learn that France has an interesting twist to their electoral rules on how many candidates run in legislative and cantonal elections. Legislative and cantonal elections in France are held under a two-round majority system. **The second round is held one week following the first one. The two candidates with the largest number of votes in the first round are automatically qualified for the second round. Yet, if another candidate receives a vote share higher than the qualifying threshold, they also qualify to the second round.** Importantly, the vote share for this threshold is calculated based on the number of registered voters. For legislative elections, the qualifying threshold is 12.5%. In cantonal elections, it is 10%, except in 2011 for which it was 12.5%. You also obtain official results of all French legislative and cantonal elections between 1978 and 2012, including close to 14,000 electoral district races. If a candidate receives a number of votes larger than 50% of the total number of valid votes in the first round, and if this number is higher than 25% of the total registered voters in the district, they are elected and there is no second round. These elections are excluded from this data set. The dataset, `france.dta`, is available on the course website (see the last page for variable descriptions. Do reach out if you have any questions about this).

1. As a first step, you should get to know the data: i) What is the unit of observation (a row) in the dataset? ii) What types of elections are covered in the data? iii) How many candidates are there in the first and second round on average? iv) What is the average turnout and null and blank votes in the second round if there are two candidates vs three candidates? What would you conclude from that? v) How many third ranked candidates in the first round pass the qualifying threshold? How many of those run in the second round? (*Hint: Use the variables threshold and threshold_party_can3.)*

2. What causal inference method would you use to study this question? Write out the model specification, then state and discuss the identification assumption in this case.

3. Produce two (descriptive) plots and discuss what these plots tell you about the data in the context of your research strategy.

4. How would you test the identification assumption? Choose at least one way using the data provided to empirically test that your choice of the research design is internally valid.

5. Estimate the effect of a third candidate on the two main outcomes: (i) voter turnout & (ii) null and blank votes. Operationalise the main treatment variables and the running variable as you see appropriate. Interpret the results, both in terms of the size of the estimates their statistical significance. Write an overall conclusion addressing your research question.

variable name	type	format	label	variable label
year	int	%10.0g		
id_canton	byte	%8.0g		
departement	str29	%29s		
election	str8	%9s		**type of election**
party_can1	str14	%14s		**party of first ranked candidate**
voteshare_t1_~1	long	%8.0g		**total votes in round 1 of first ranked candidate**
voteshare_t2_~1	long	%10.0g		**total votes in round 2 of first ranked candidate**
ran_t2_can1	float	%9.0g		**whether first ranked candidate ran in round 2**
ideology_can1	float	%9.0g		**ideology of first ranked candidated**
party_can2	str14	%14s		**party of second ranked candidate**
voteshare_t1_~2	long	%8.0g		**total votes in round 1 of second ranked candidate**
voteshare_t2_~2	long	%10.0g		**total votes in round 2 of second ranked candidate**
ran_t2_can2	float	%9.0g		**whether second ranked candidate ran in round 2**
ideology_can2	float	%9.0g		**ideology of second ranked candidated**
party_can3	str14	%14s		**party of third ranked candidate**
voteshare_t1_~3	long	%8.0g		**total votes in round 1 of third ranked candidate**
voteshare_t2_~3	long	%10.0g		**total votes in round 2 of third ranked candidate**
threshold_par~3	float	%9.0g		**qualifying share (votes/registered voters) for third candidate in first round**
ran_t2_can3	float	%9.0g		**whether third ranked candidate ran in round 2**
ideology_can3	float	%9.0g		**ideology of third ranked candidated**
inscrits_t1	long	%12.0g		**Registered voters, 1st round**
parties_t1	float	%9.0g		**Number of candidates, 1st round**
elected_t1	float	%9.0g		**Elected in first round**
parties_t2	float	%9.0g		**Number of candidates, 2nd round**
threshold	float	%9.0g		**qualifying threshold for third ranked candidate**
turnout_t1	float	%9.0g		**Turnout, 1st round**
cvotes_t1	float	%9.0g		**Valid Turnout, 1st round**
blancsnull_t1	float	%9.0g		**Null/Blank Votes, 1st round**
turnout_t2	float	%9.0g		**Turnout (2nd)**
cvotes_t2	float	%9.0g		**Valid Turnout (2nd)**
blancsnull_t2	float	%9.0g		**Null/Blank Votes (2nd)**
nosecond	float	%9.0g		**No second round, 3 criteria**
margin_t2	float	%9.0g		**Voteshare difference 1st-2nd cand**
margin_t1	float	%9.0g		**Voteshare difference 1st-2nd cand**
threecand	float	%9.0g		**Three candidates in second round**

Figure 1: Variables in the Data and their Labels